PARTS LIST

Item	Qty	REFERENCE DESIGNATOR	PART NUMBER	Description	MANUFACTURER
1	1		S1101EFF1553	BARE BOARD	ZIN TECHNOLOGIES
2	1	C1	C0805C104K1RACTU	CAPACITOR 0.1uF 100V CERAMIC 10% 0805 SMD	KEMET
3	1	C2	T494B106K016AT	CAPACITOR 10.0uF 16V TANTALUM 10% B-SIZE SMD	KEMET
4	2	C3, C4	C2012X7S2A105K125AB	CAPACITOR SMD, CERAMIC, 1uF, 100V, ±10%, X7S, 0805	TDK
5	1	R1	ERJ-6ENF1502V	RESISTOR, 15K, 1/8W, 1%, 0805	PANASONIC - ECG
6	1	U1	AD8628ARTZ	SINGLE SUPPLY OP AMP, SOT-23	ANALOG DEVICES
7	1	U2	XP-0249	THERMOPILE DETECTOR, THIN FILM BASED	DEXTER RESEARCH CENTER
8	1	U3	LT1790ACS6-1.25	LOW DROPOUT REFERENCE SOURCE	LINEAR TECHNOLOGY
9	1	U4	LM20CIM7	TEMPERATURE SENSOR, 2.4V, 10uA, SC70	NATIONAL
10	A/R		TZE-211	LABEL, 1/4 BLK LETTERS WHT BACKROUND	BROTHER
11	A/R		3140	RTV FOR CONFORMAL COATING	DOW CORNING
12	A/R		PR-1204	PRIMER FOR CONFORMAL COATING	DOW CORNING
13	A/R		SN63PB37/RMA	SOLDER	
14	2	R2, R3	ERJ-6ENF15R0V	RESISTOR 15.0 OHM 1/8W 1% 0805 SMD	PANASONIC - ECG

7. CONFORMAL COATING, STAKING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.1 AND THE FOLLOWING NOTES:
   A. PERFORM CLEANLINESS TEST.
   B. BAKE PWA AT 93°C ±5.5°C (200°F ±10°F) FOR 4 HOURS PRIOR TO CONFORMAL COATING.
   C. MASK OFF ALL DESIGNATED AREAS USING KAPTON TAPE. ALL MASKING OPERATIONS SHALL BE PERFORMED UNDER AN AIR IONIZER.
   D. MEASURE CONFORMAL COATING THICKNESS AT DESIGNATED AREAS.
   E. CONFORMAL COAT THE COMPLETE ASSEMBLY USING ITEM 11 AND ITEM 12.
8. PLACE THE COMPLETED ASSEMBLY IN CLEAN, STATIC DISSIPATIVE PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER PER SAE AS478, METHOD 35. STORE IN A PROJECT APPROVED STORAGE AREA.
9. REFERENCE SCHEMATIC: S1101EFS1553
10. EXPEDITION CONTROL FILES: S1101EFA1553_REV_B.zip
11. PLACE ITEM 7 (U2) DIRECTLY ON TOP OF ITEM 9 (U4) WITH THE BOTTOM OF ITEM 7 SLIGHTLY BELOW THE TOP OF ITEM 9.

NOTES: UNLESS OTHERWISE SPECIFIED

1. CERTIFICATE OF CONFORMANCE REQUIRED FOR ALL COMMERCIAL PARTS.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
4. DEMOISTURIZE (BAKE OUT) PWB (ITEM 1) FOR A MINIMUM OF 2 HOURS AT 93°C ±5.5°C (200°F ±10°F.) PERFORM SOLDERING IMMEDIATELY AFTER BAKING, OR STORE IN A CLEAN, DRY, SEALED STATIC DISSIPATIVE BAG OR CONTAINER.
5. MARK THE ASSEMBLY WITH REV AND S/N USING ITEM 10.
6. ASSEMBLY, SOLDERING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.2, NASA-STD-8739.3, AND THE FOLLOWING NOTES:
   A. INSTALL AND SOLDER ALL SURFACE MOUNT (SMT) PART BEFORE THROUGH HOLE PARTS USING SOLDER (ITEM 13.)
   B. FOR THROUGH HOLE PARTS, LEAD TERMINATIONS SHALL BE STRAIGHT THROUGH (NOT LAPPED OR CLINCHED.)

REV	DESCRIPTION	APPROVED
B	REVISE PER ACME-ECO-392.	MH 5/5/15 TG 5/6/15 DS 5/13/15
A	REVISE PER ACME-ECO-338 AND ACME-ECO-352	TAG 9/11/14 DAS 9/11/14 RJP 9/11/14

REVISION HISTORY

.

S1101MFA1550

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCES: ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.

APPROVED: E. McCLOSKEY 5/13/15

PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	R. PFEIFFER 4/23/14	QUALITY	D. CHOVAN 4/25/14
REVIEWER	D. STRANG 4/24/14	MATERIALS	M. WILKINSON 4/23/14
MECHANICAL	A. DRAKE 4/24/14	PROJ LEAD	B. BOROWSKI 4/28/14
ELECTRICAL	T. GOBELI 4/24/14	-	-
STRUCTURAL	C. WERNER 4/25/14	-	-
SAFETY	W. BORRELLI 4/23/14	-	-

DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994

CONTRACT NUMBER: NNC13ZMA015R
DELIVERY CODE: S2-BT3

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 625-2200
WWW.ZIN-TECH.COM

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE

ACME
BRE RADIOMETER BOARD
ASSEMBLY DRAWING

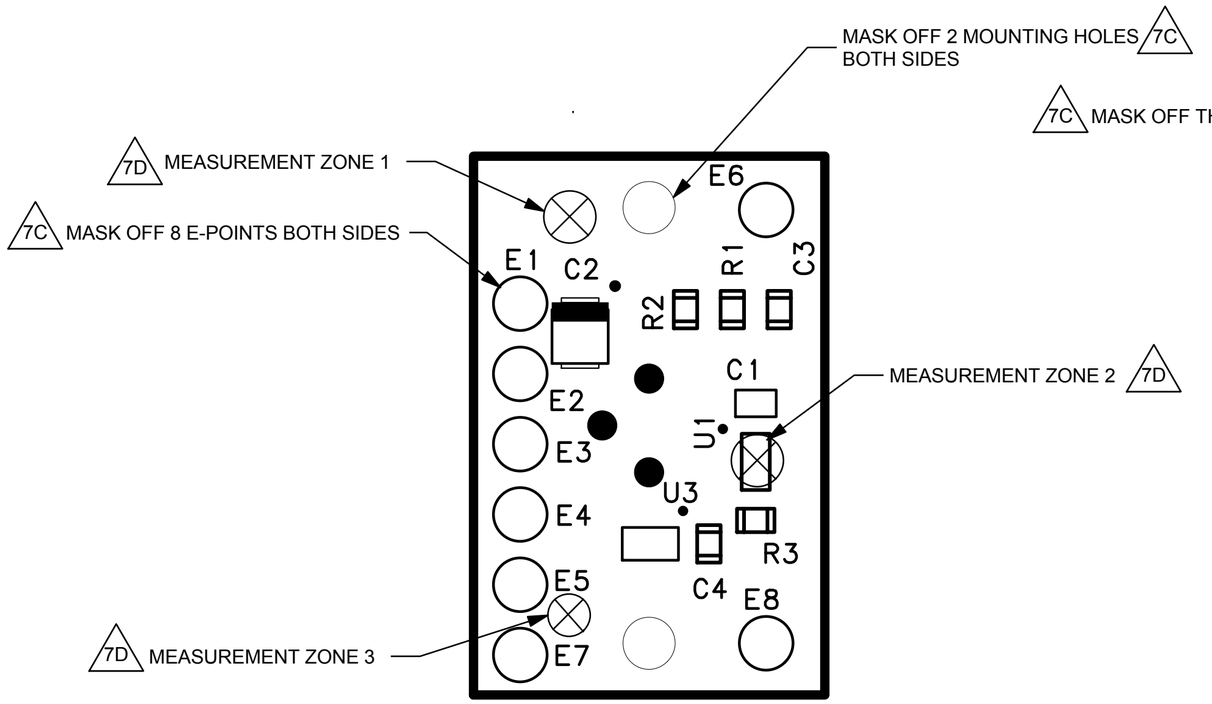
SIZE	CAGE CODE	DWG NO	REV
C	0SG68	S1101EFA1553	B

SCALE: NONE | INITIAL REL DATE 4/28/14 | SHEET 1 OF 2

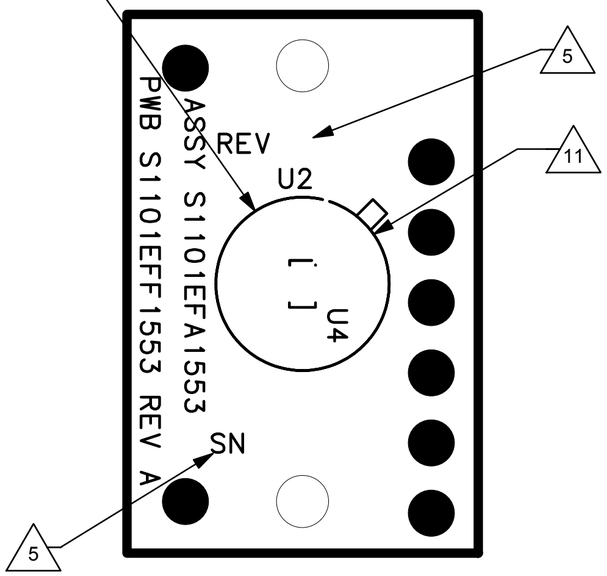
RELEASE LEVEL: .
MODEL NAME: S1101EFA1553_REV_B.bpd

BLUEPRINT
VERSION 3.0

DWG NO S1101EFA1553 SH 1 REV B



PRIMARY SIDE VIEW SCALE 4X



SECONDARY SIDE VIEW SCALE 4X

SIZE	CAGE CODE	DWG NO	REV
C	0SG68	S1101EFA1553	B
SCALE: NONE	INITIAL REL DATE 4/28/14	SHEET 2 OF 2	

BLUEPRINT
VERSION 3.0

REV B
SH 2
DWG NO S1101EFA1553